The anatomy of emotion

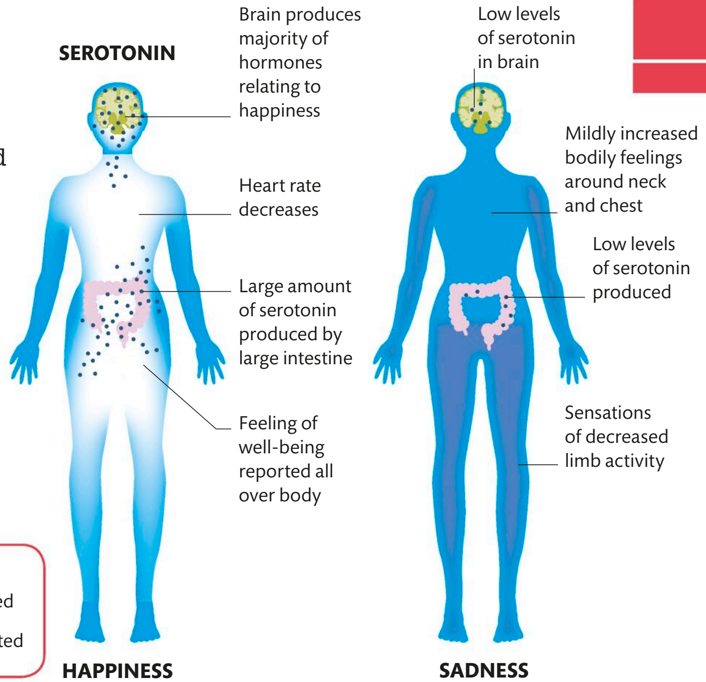
In response to a stimulus, the brain initiates hormonal changes that, in turn, trigger physiological changes that prime us to respond in appropriate ways to the current emotional state. Heart rate changes, altered blood flow to the muscles, and sweating are associated with heightened emotions. These changes can be felt consciously, increasing the emotional intensity.

WHAT IS THE PURPOSE OF LAUGHTER?

The relaxation that results from a bout of laughter inhibits the biological fight-or-flight response.

Happiness and sadness
Serotonin, dopamine, oxytocin, and endorphins are hormones that affect our happiness profoundly. Emotions are felt across the body, with different emotions felt in different places. The effects of serotonin are shown here.

KEY
Positive feelings reported
Negative feelings reported



Unconscious emotions

For primitive automatic responses, such as the fight-or-flight reflex, speed is critical. Emotionally charged stimuli presented too fast to be consciously perceived can evoke emotional responses and activate the amygdala. These initial responses shape how the cortex processes information. The amygdala is involved in emotional memory that may be automatically activated in the future.

Sensory cortex
Sensory information transmitted to the sensory cortex is extensively processed toward conscious perception and integrated with stored information. This takes time.

Hippocampus
The hippocampus processes consciously perceived information to form memories. It also compares incoming signals to previous memories to adjust emotional responses.

SLOW AND ACCURATE ROUTE

Thalamus
Incoming information is relayed to the amygdala for quick assessment and action and also to cortical areas where it enters conscious awareness.

QUICK AND DIRTY ROUTE

Amygdala
The amygdala instantly assesses the emotional importance of incoming information content and rapidly sends signals to other areas for immediate bodily action.

Hypothalamus
Signals from the amygdala trigger hormonal changes and output to the autonomic nervous system to prime the body to respond to emotional stimuli.

Two routes
Conscious processing of emotions involves integrating sensory information with stored memories and reasoned evaluations of a situation—this is the "slow and accurate route." In contrast, unconscious responses, via the "quick and dirty route," happen much faster. The prefrontal cortex is important in conscious emotional regulation.